

Electrifying boda-boda transportation with battery-swapping: a techno-economic and grid-aware rollout for Kampala Metropolis

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ABSTRACT

Electrifying motorcycle taxi (boda-boda) services offers a significant opportunity to reduce emissions and petroleum dependence in rapidly growing African cities. Still, large-scale deployment of electric two-wheelers (E2Ws) can stress distribution networks if infrastructure planning is not grid-aware. This study develops an integrated techno-economic and policy framework to evaluate battery-swapping-based electrification of boda-boda fleets in the Greater Kampala Metropolitan Area (GKMA). GPS-derived travel-demand data are used to characterize rider duty cycles and temporal energy demand, which inform a geospatial optimization of battery-swapping and charging hubs constrained by feeder routes. Comparative total cost-of-ownership analysis and feeder-level power-flow simulations assess infrastructure performance under alternative tariff and demand-side management regimes. A randomized controlled trial evaluates financing schemes for operators, while an energy-oriented input–output model linked to the KAMPALA-TIMES transport module estimates macroeconomic ripple effects. Results show that hybrid swap–charge configurations reduce ownership costs by 18–27%, maintain feeder utilization below 85% of design capacity, and deliver measurable reductions in energy and carbon intensities of GDP. The study provides a grid-compatible, investment-ready roadmap for the deployment of electric motorcycles in Sub-Saharan African cities.

Introduction

Motorcycle taxis, locally known as boda-bodas, have become the backbone of urban mobility across Sub-Saharan Africa, offering flexible, low-cost transport for millions of commuters while supporting the livelihoods of thousands of operators [1]. In Uganda's Greater Kampala Metropolitan Area (GKMA), they account for over 60% of intra-urban passenger trips, effectively serving as the last-mile link in the city's transport ecosystem [2]. However, the sector's heavy reliance on imported petroleum fuels has led to high operating costs, urban air pollution, and an increasing share of greenhouse gas (GHG) emissions [3]. The combustion-based boda-boda fleet is estimated to contribute nearly 40% of transport-related particulate emissions in Kampala [4], exacerbating respiratory health risks and undermining the attainment of SDG 3.6 (reducing road injuries and deaths) and SDG 13 (climate action).

The electrification of two-wheelers, primarily through battery-swapping and shared charging models, has become a strategic

approach to reducing carbon emissions in this critical yet informal transportation sector [5,6]. Several countries, including Kenya, Rwanda, and India, are testing e-motorcycle transitions that combine local assembly with digital payment systems, battery leasing, and grid-responsive charging [7,8]. However, scaling such initiatives in African cities faces systemic problems, mainly (i) the lack of optimized siting frameworks for swap and charge hubs, (ii) limited consideration of distribution grid constraints in planning, and (iii) financial challenges for riders due to high upfront costs and unstructured tariffs [9,10].

Existing studies on e-mobility in developing contexts mainly focus on vehicle performance or life-cycle emissions, often ignoring the cross-sectoral interactions between techno-economics, grid stability, and finance-policy ecosystems [11–13]. Additionally, while regional initiatives like the Africa Electric Mobility Framework and Uganda's NDC Implementation Plan highlight the potential for e-mobility adoption, there is limited empirical evidence on the spatially optimized, grid-aware deployment of e-boda infrastructure [14]. This knowledge gap restricts policymakers and investors from reducing risks in e-boda

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business models and maintaining long-term system sustainability.

Electrification of motorcycle taxi systems in African Cities: evidence from regional pilots

Across Sub-Saharan Africa, motorcycle taxis, commonly known as boda-bodas, okadas, or moto-taxis, have become a dominant form of informal yet indispensable urban mobility [15]. In recent years, these systems have become a focal point of transport electrification efforts due to their high utilization, predictable duty cycles, and disproportionate contribution to urban air pollution and petroleum demand [4]. Unlike private passenger vehicles, motorcycle taxis operate as income-generating assets, making their electrification a techno-economic and infrastructure-planning challenge rather than a purely consumer adoption problem [8].

Empirical evidence from East and Southern African cities shows that electric motorcycle (e-motorcycle) deployment has moved beyond pilot demonstrations into early-stage commercialization, particularly where battery swapping, digital payments, and cooperative financing are integrated [8]. In Nairobi and other Kenyan cities, multiple operators have deployed e-motorcycles at scale, relying predominantly on battery-as-a-service and swapping models to mitigate range anxiety and minimize rider downtime [16]. These initiatives have shown that swapping systems are operationally advantageous under high daily mileage (80–120 km/day), but have also revealed emerging challenges related to station density, feeder congestion, and tariff exposure under unmanaged charging regimes [17,18].

In Kigali, Rwanda's strong regulatory coordination and compact urban form have enabled structured e-motorcycle rollouts supported by standardized batteries, centrally managed swap stations, and growing use of solar-assisted charging [19]. Kigali's experience illustrates how policy coherence, import facilitation, and early standard-setting can accelerate adoption [20]. However, even in this relatively controlled context, recent analyses highlight that grid constraints, land availability for hubs, and long-term battery lifecycle management remain unresolved engineering questions, particularly as penetration increases beyond pilot scale [21].

In Uganda, early e-motorcycle pilots and emerging private-sector deployments have confirmed the sector's potential while underscoring structural constraints tied to highly dispersed travel patterns, heterogeneous road quality, and feeder-level grid limitations within the Greater Kampala Metropolitan Area (GKMA) [8]. Kampala riders typically combine passenger transport with courier and parcel services, resulting in longer daily distances and more irregular temporal demand than in Kigali [22]. These characteristics intensify energy throughput requirements, strengthen the economic case for hybrid swap-charge systems, and increase the importance of grid-aware siting and demand-side management [23,24].

Although focused more broadly on electric mobility and grid integration, evidence from Cape Town and other South African metros provides a relevant African comparator for distribution-level constraints and managed charging strategies [25,26]. Modeling and pilot studies in Cape Town consistently show that unmanaged clustering of EV charging, particularly in dense commercial corridors, can rapidly strain low- and medium-voltage feeders, even in comparatively robust power systems [27,28]. These findings reinforce the need for feeder-constrained planning, time-of-use responsiveness, and spatial optimization when extending electrification to high-utilization vehicle classes, such as motorcycle taxis [29].

Taken together, African literature converges on several key insights. First, battery swapping is emerging as the dominant operational model for motorcycle taxi electrification under high-duty-cycle conditions, but its scalability depends critically on spatial planning and grid compatibility [30]. Second, grid impacts are highly localized, requiring feeder-level analysis rather than system-average assumptions [21,31]. Third, financial viability and adoption are closely tied to asset

ownership structures, cooperative guarantees, and daily cash flow stability, rather than headline vehicle prices [32]. Finally, while African pilots demonstrate technical feasibility, there remains a clear gap in integrated, grid-aware techno-economic frameworks that link mobility demand, infrastructure siting, power-flow constraints, financing design, and economy-wide impacts within a single analytical architecture [33]. Recent Kampala-specific mobility system analyses [34] and Nairobi grid-impact studies [28] further underscore the need for integrated mobility-grid planning frameworks rather than vehicle-centric electrification strategies.

This study directly addresses this gap by situating Kampala's boda-boda electrification pathway within the evolving African evidence base and extending it through a feeder-constrained, demand-responsive, and policy-linked engineering framework. By explicitly benchmarking Kampala against Nairobi, Kigali, and Cape Town, the analysis moves beyond isolated case studies. It provides system-level insights relevant to African cities experiencing rapid growth in informal mobility amid constrained grid and fiscal conditions. Table 1 synthesizes findings from peer-reviewed studies, policy analyses, and documented African pilot deployments to contextualize the Kampala case within regional electrified motorcycle and broader e-mobility transitions.

The regional African evidence reviewed above shows that electric motorcycle transitions are no longer speculative but are unfolding through diverse operational, infrastructural, and institutional pathways. However, existing African studies and pilots remain largely fragmented—typically examining vehicle performance, financing models, or policy enablers in isolation—without systematically integrating mobility demand, distribution-grid constraints, infrastructure siting, and economy-wide implications into a single analytical framework. This fragmentation limits the transferability of lessons across cities and constrains policymakers, utilities, and investors from evaluating electrification pathways at scale. Building on this gap, the present study advances an integrated, grid-aware, and policy-linked engineering framework that consolidates these dimensions into a unified decision-support architecture tailored to high-utilization motorcycle taxi systems in African cities.

Motivated by Uganda's Vision 2040, National Development Plan IV, and the 2023 Energy Transition Plan [35], this study develops an integrated modeling approach that combines bottom-up technical optimization with macroeconomic policy assessment to inform evidence-based electrification strategies for Kampala's boda-boda ecosystem. The KAMPALA-TIMES transportation module is linked to an energy-oriented input-output framework to evaluate job creation, income effects, the energy intensity of GDP, and carbon abatement potential, bridging engineering rigor and policy insight.

Accordingly, the study is guided by four overarching research questions:

1. What configuration of charging and swapping minimizes the total cost of ownership (TCO) while ensuring operational reliability for boda-boda operators?
2. Where should swap and charging hubs be spatially located to balance rider demand density with the technical limits of local distribution feeders?
3. Which tariff regimes and financing structures (e.g., lease-to-own, pay-as-you-ride) enable cash-flow-positive operations and equitable sector participation?
4. How can the energy-oriented input-output modeling framework be applied to capture the macroeconomic feedbacks, such as employment, household income, energy intensity of GDP, and carbon intensity of GDP effects arising from large-scale electrification of the boda-boda sector in Kampala Metropolitan?

By addressing these queries, the study contributes to the emerging discourse on data-driven e-mobility transitions in low and middle-income countries (LMICs). It offers a replicable framework for

Table 1
Comparative Evidence from African Electric Motorcycle and E-Mobility Pilots.

City / Country	Dominant Motorcycle Taxi Model	Electrification Approach	Infrastructure Model	Grid & Energy Considerations	Key Lessons for This Study	Representative Literature
Nairobi, Kenya	High-utilization motorcycle taxis (passenger + courier)	Rapid e-motorcycle deployment	Battery swapping (BaaS), digital PAYGo	Localized feeder congestion emerging; tariff exposure without DSM	Swapping minimizes downtime but requires grid-aware station density	Martin et al. [8], Simwaba & Qutieshat [16], Mbugua et al. [10]
Kigali, Rwanda	Regulated motorcycle taxi system	Structured e-motorcycle rollout	Standardized batteries, centralized swap hubs	Increasing use of solar-assisted hubs; land constraints	Policy coherence accelerates adoption, but grid constraints persist	Furaha et al. [19], Martin et al. [8]
Kampala, Uganda	Highly informal, mixed passenger–cargo use	Early-stage electrification pilots	Emerging hybrid swap–charge concepts	Heterogeneous feeders, variable road quality, longer duty cycles	Requires hybrid infrastructure and feeder-constrained siting	Kimuli et al. (2024, [3], 2026)
Cape Town, South Africa	Formal transport and private EV focus	EV integration studies and pilots	Managed charging emphasis	Demonstrated LV/MV feeder sensitivity to clustered charging	Reinforces need for DSM and spatial optimization	Nnene et al. [20]; TRD modeling studies

designing grid-compatible, economically viable, and socially inclusive electrification pathways for urban two-wheeler transport systems.

Contribution and novelty statement

This study develops an integrated, grid-aware techno-economic framework to assess the electrification of motorcycle taxi (boda-boda) fleets in Sub-Saharan African cities. Unlike existing studies that typically examine vehicle performance, charging infrastructure, or financing mechanisms in isolation, the proposed approach integrates travel-demand estimation, distribution-level power-flow analysis, geospatial infrastructure siting, and techno-economic evaluation into a unified energy-systems modeling architecture.

The first contribution is the explicit incorporation of feeder-level distribution grid constraints into the spatial optimization of battery-swapping and charging hubs, enabling infrastructure planning that aligns with local network capacity and operational limits. This departs from conventional e-mobility planning approaches that rely on system-average or unconstrained grid assumptions. The second contribution is coupling the KAMPALA-TIMES transportation module with an energy-oriented input–output framework, enabling analysis that extends beyond project-level metrics and quantifies macroeconomic ripple effects, including employment impacts and changes in the energy and carbon intensities of gross domestic product. The third contribution is the use of a pilot-scale randomized controlled trial to empirically examine the feasibility of alternative financing schemes for informal transport operators, thereby validating key techno-economic assumptions rather than population-level adoption forecasts.

Together, these contributions advance a reproducible and policy-relevant analytical framework for grid-compatible electric motorcycle deployment, with direct applicability to urban contexts characterized by high informal mobility demand and constrained distribution infrastructure. The framework supports infrastructure planning, tariff design, and investment appraisal aligned with national development strategies, climate commitments, and sustainable development objectives.

Assumptions and limitations of the study

The analytical framework applied in this study relies on a set of technical, economic, and behavioral assumptions that enable tractable modeling of e-boda electrification within the complex energy–mobility system of the Greater Kampala Metropolitan Area (GKMA). These assumptions are intentionally constrained to maintain transparency, facilitate reproducibility, and reflect the data environment typical of rapidly urbanizing African cities.

On the technical side, feeder-level distribution data from UMEME/ERA/UEDCL are assumed to represent 2024 loading, voltage, and transformer conditions accurately; informal connections and unreported

technical losses are not included in the power-flow simulations. Hub operations follow standardized lithium-ion performance profiles, with typical charging turnover of 1.5 h and annualized battery degradation of 10–12%, consistent with regional evidence [36,37]. The model further assumes stable grid availability, given the incorporation of embedded PV–BESS systems at selected hubs.

Economic assumptions reflect 2024–2025 OEM quotations, stable import-duty policies, and prevailing ERA time-of-use tariffs (off-peak 0.14 USD/kWh; peak 0.22 USD/kWh). Financial models for lease-to-own and pay-as-you-ride schemes apply a uniform 10% interest rate over a 24-month horizon. Macroeconomic feedbacks within the energy-oriented input–output module assumes no major exogenous shocks to fuel prices or fiscal incentives.

Behaviorally, rider travel patterns are derived from 30-day GPS traces, assuming rational repeatability and a willingness-to-adopt above 65% under cost neutrality. No significant rebound effects are modeled.

Limitations include an incomplete representation of informal mobility routes, simplified outage dynamics, uncertainty in long-term battery aging, and risks to policy continuity. Although the methodological framework is adaptable, the quantitative results are tailored to GKMA conditions and need adjustments for other urban settings. These limitations do not weaken the study's conclusions but set realistic boundaries for interpreting system-wide outcomes.

Conceptual framework and theoretical underpinnings

The study is grounded in a multi-theoretical framework that integrates engineering optimization, energy-systems modeling, and economic theory to inform the electrification of Uganda's boda-boda subsector within the country's urban transport landscape [14]. This framework provides a systems-thinking lens for systematically examining the interdependencies among the techno-economic, grid-integration, and policy-finance dimensions.

At its core, the conceptual approach places e-boda electrification at the intersection of energy-systems theory, which emphasizes demand–supply balancing, grid reliability, and power-flow constraints; urban transportation economics, which focuses on cost structures, efficiency, and behavioral choice; and sustainability transition theory, which centers on institutional alignment and socio-technical change. By integrating these perspectives, the study evaluates not only the technical feasibility of charging and swapping configurations but also their economic viability, distributional effects, and compatibility with Uganda's regulatory and policy environment.

This theoretical grounding ensures the analysis captures the multi-scalar nature of two-wheeler electrification, spanning micro-level engineering considerations, *meso*-level infrastructure planning, and macro-level policy and economic impacts.

Energy systems and technology diffusion theories

Energy systems theory is the first pillar of the framework, emphasizing that urban energy transitions arise from interactions between supply-side infrastructure and demand-side technology adoption [7,38–47]. In this view, e-boda electrification is not only an engineering optimization task that reduces operating costs, power losses, and grid stress but also a socio-technical reorganization of energy flows, mobility practices, and institutional structures [48,49].

To understand adoption dynamics, the study employs Rogers' Diffusion of Innovations Theory [50], which explains how new technologies spread through a social system. Key attributes, including relative advantage, compatibility, complexity, trialability, and observability, shape perceived benefits for riders and are operationalized using behavioral evidence from the RCT. This synthesis enables a rigorous assessment of how financing structures and operational performance influence diffusion trajectories across the Greater Kampala Metropolitan Area (GKMA).

Techno-economic and systems optimization framework

The study operationalizes its engineering foundation through a techno-economic optimization framework grounded in bottom-up energy-systems modeling, comparable to that used in the TIMES-VEDA platform [51]. The framework integrates GPS-derived travel-demand modeling to quantify temporal and spatial energy needs, multi-objective geospatial siting optimization to place charging and swapping hubs within rider-demand clusters while respecting feeder-capacity limits, and a Total Cost of Ownership (TCO) assessment that captures lifecycle cost variations across tariff regimes, financing structures, and battery-leasing plans. This integrated analytical layer identifies the least-cost, grid-compatible infrastructure configuration that meets ERA's distribution system requirements and KCCA's urban mobility zoning guidelines.

Grid integration and power flow theory

Grid integration is examined using power-flow and network-reliability theory to assess how rising e-boda charging demand interacts with distribution-system stability [52,53]. Using Kirchhoff-based load-flow optimization [54], the study evaluates voltage deviations,

transformer loading, and congestion risks across feeder configurations. Smart-charging and battery-swapping schedules are integrated into a demand-side management (DSM) framework [12], enabling peak-load shifting and improving feeder utilization. This approach ensures that e-mobility expansion remains compatible with Uganda's grid reliability and operational safety objectives.

Policy, finance, and institutional theories

The policy and institutional aspects of e-boda electrification are examined through Institutional Theory [55,56] and the Policy Coherence for Sustainable Development (PCSD) framework [57,58], highlighting that technological transitions succeed only when regulatory mandates across transport, energy, finance, and the environment are aligned. The policy–finance interface draws on Public–Private Partnership (PPP) theory [59] to show how concessional credit, tariff reform, and results-based financing encourage private-sector investment. Financial innovation via PAYGo, micro-leasing, and SACCO-based credit is analyzed using Behavioral Economics, specifically Prospect Theory [60,61], which explains rider risk perceptions and adoption behavior under repayment uncertainty.

The integrated conceptual framework

Fig. 2 presents the integrated conceptual framework guiding this inquiry, highlighting interactions among the techno-economic, grid-integration, and policy-finance dimensions that shape e-boda electrification in the Kampala Metropolis. The techno-economic layer evaluates engineering performance, geospatial hub siting, and lifecycle cost metrics to identify feasible charging and swapping configurations. The grid dimension incorporates feeder-capacity limits and power-flow behavior to ensure that electrification aligns with Uganda's reliability and resilience objectives. The policy-finance dimension captures tariff design, concessional credit, and results-based financing mechanisms that enable market scalability and inclusion for small operators.

These dimensions converge in a decision-support nexus that synthesizes technical, economic, and institutional insights to deliver system-level outcomes, including emissions reductions, job creation, and grid stability, consistent with Vision 2040, Uganda's NDCs, and the SDGs.

ELECTRIFYING BODA-BODA FLEETS WITH BATTERY-SWAPPING

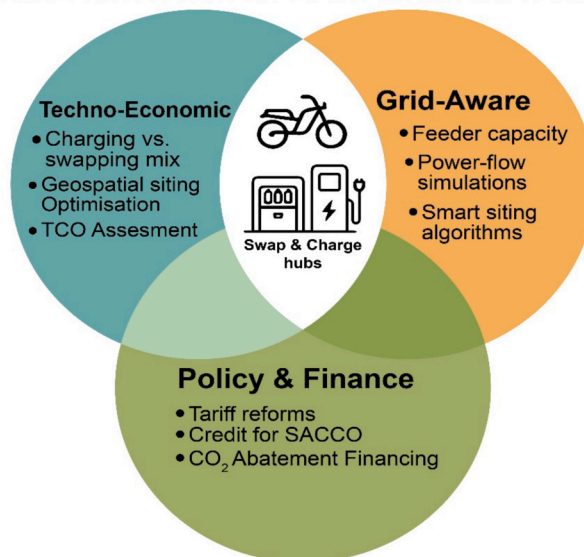


Fig. 1. Graphical Abstract.

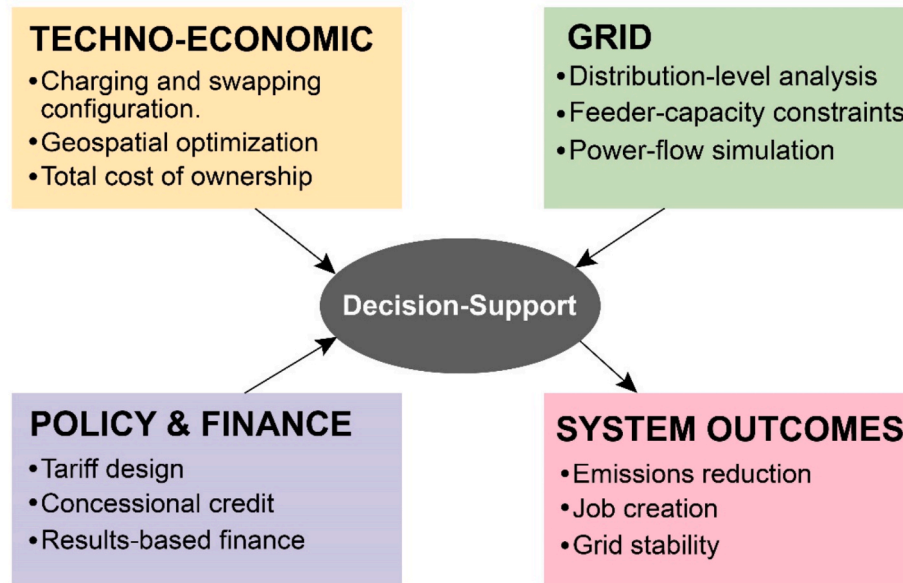


Fig. 2. Conceptual Framework.

The framework advances theory by linking engineering systems modeling and development economics, demonstrating that energy transition models can serve as decision-making platforms for inclusive transport electrification in low- and middle-income countries (LMICs). It broadens the Energy-Economy-Environment (E3) paradigm by integrating mobility behavior and financial innovation, presenting a new E3-Mobility-Policy (E3MP) framework for urban e-mobility shifts in data-scarce settings.

Methodology

The methodological design of this study employs a multi-layered modeling framework that integrates engineering optimization, financial experimentation, and macroeconomic policy analysis. This approach enables a comprehensive evaluation of the electrification of Kampala's boda-boda sector from both technical and macroeconomic perspectives. Fig. 3 illustrates the sequential analytical stages, including (i) travel-demand modeling, (ii) geospatial siting optimization, (iii) techno-

economic analysis, (iv) distribution grid simulation, (v) financial pilot testing, and (vi) macroeconomic assessment. Each methodological component is detailed in the subsections that follow.

Travel-demand and vehicle energy modeling

The first analytical stage involves travel-demand modeling to characterize the operational behavior and associated energy use of conventional boda-boda fleets across the Greater Kampala Metropolitan Area (GKMA). A sample of approximately 300–500 motorcycles was equipped with anonymized GPS tracking devices that recorded high-frequency spatiotemporal data over a continuous 30-day observation period. These data capture route choice, trip chaining, dwell times, and daily distance traveled, thereby reflecting the heterogeneous and high-utilization nature of informal motorcycle taxi operations in Kampala.

The GPS records were processed using Python-based geospatial and data analysis libraries, including pandas, geopandas, and OSMnx, to reconstruct individual trips and derive trip start–end locations, distance distributions, temporal activity profiles, and spatial demand density. These outputs provide an empirical basis for estimating rider duty cycles and for aggregating demand across space and time. The resulting demand density maps form the spatial foundation for subsequent infrastructure siting and grid-impact assessment.

Vehicle energy consumption is not treated as a fixed input in the analysis. Instead, per-kilometer electricity demand for electric motorcycles is treated as an emergent outcome of interactions among physical and operational factors that characterize real-world riding conditions in GKMA[62,63]. Energy use is modeled as a function of combined rider and passenger mass, additional cargo payload, rolling resistance associated with heterogeneous pavement quality, aerodynamic drag determined by operating speed, localized road gradients, and stop-and-go driving patterns arising from congestion and frequent acceleration events[64–66]. These components are represented using a standard tractive energy formulation that decomposes total energy demand into rolling resistance, aerodynamic, inertial, and grade-related contributions, consistent with established approaches in electric two-wheeler and light-duty electric vehicle energy modeling [67–69].

Empirical energy-intensity values reported for East African electric-motorcycle pilots, typically 40–50 Wh km^{−1}, are therefore used as central reference points rather than deterministic assumptions[70]. To reflect operational variability, a structured sensitivity analysis is

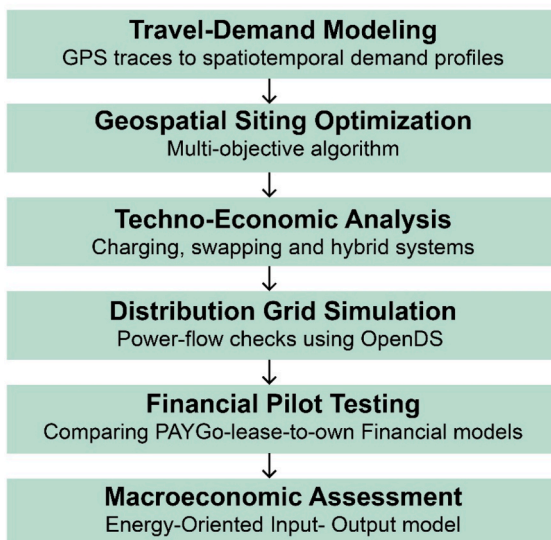


Fig. 3. Methodological Framework.

conducted across realistic ranges of rider mass, payload, rolling resistance, and road gradient, yielding an effective energy-intensity envelope of approximately 35–65 Wh km⁻¹ under GKMA conditions [71,72]. The resulting distributions of daily electricity demand are then propagated into infrastructure sizing, techno-economic evaluation, and feeder-level grid simulations. In this way, mobility behavior, physical operating conditions, and energy demand are consistently linked within the integrated modeling framework, ensuring that downstream analyses are grounded in empirically observed travel patterns and realistic vehicle performance characteristics.

To ensure the reliability of distance estimates and derived energy-demand metrics, GPS data acquisition and processing procedures were explicitly designed to control measurement error and data gaps. Motorcycle trajectories were recorded at 5–10 s intervals using anonymized GPS loggers, enabling resolution of stop-go dynamics and route choice in congested urban conditions. Raw point streams were segmented into individual trips using dwell-time and speed thresholds, then map-matched to the OpenStreetMap road network with a Hidden Markov Model-based algorithm to correct positional drift in dense corridors and intersections. Linear interpolation was applied only for short signal gaps (<30 s), while trips with prolonged signal loss or implausible spatial jumps were excluded rather than imputed. Additional quality-control filters removed outliers based on speed and spatial consistency, retaining approximately 85–90% of recorded trips. Robustness checks indicate that residual GPS uncertainty introduces less than 3–5% variation in aggregate daily distance estimates, well within the energy-intensity sensitivity bounds considered in the analysis, confirming that downstream energy, grid, and cost results are not materially sensitive to plausible GPS measurement errors.

Geospatial siting optimization

The second methodological stage applies a multi-objective geospatial optimization model to determine optimal locations for charging and battery-swapping hubs across the Greater Kampala Metropolitan Area (GKMA). The problem is formulated as a p-median, capacity-constrained siting model, jointly minimizing (i) average rider-hub distance and (ii) distribution-feeder loading impacts.

Decision variables

- $x_j \in \{0, 1\}$: 1 if a hub is sited at the candidate location j , 0 otherwise.
- $y_{ij} \in [0, 1]$: proportion of demand node i assigned to the hub j .

Objective function

A weighted bi-objective function is defined as:

$$\min Z = \alpha \sum_{i \in I} \sum_{j \in J} d_{ij} y_{ij} + (1 - \alpha) \sum_{j \in J} \frac{L_j(x_j)}{C_j}$$

Where:

- d_{ij} = travel distance between demand node i and hub j ,
- $L_j(x_j)$ = projected additional feeder load at node j if a hub is installed,
- C_j = available feeder capacity margin,
- $\alpha \in [0, 1]$ = trade-off weight between accessibility and grid feasibility.

Constraints

1. Demand Assignment Constraint

$$\sum_{j \in J} y_{ij} = 1, \forall i \in I$$

2. Hub Activation Constraint

$$y_{ij} \leq x_j, \forall i \in I, j \in J$$

3. Feeder capacity constraint

$$L_j(x_j) \leq C_j, \forall j \in J$$

4. Zoning and Land-Use Eligibility Constraint

$$x_j = 0 \text{ if } j \notin Z$$

where Z denotes KCCA-permitted zones and sites not in violation of land-use ordinances.

5. Hub Count Constraint

$$\sum_{j \in J} x_j = p$$

With p varying across deployment scenarios.

Implementation and scenario design

The optimization was implemented in ArcGIS Pro using weighted overlay layers and a p-median solver. Feeder-capacity maps were integrated from ERA's 2024 distribution datasets; zoning and land-use constraints were incorporated from KCCA geospatial layers.

Three siting scenarios (swap-dominant, charge-dominant, and hybrid) were modeled using different values and accessibility trade-off weights, producing location sets that balance demand density with feeder resilience[73,74].

Techno-economic analysis

A Total Cost of Ownership (TCO) framework was developed to evaluate the life-cycle cost performance of alternative infrastructure and operational configurations, including battery swapping, plug-in charging, and hybrid swap-charge deployments[75,76]. The analysis translates engineering outputs, such as energy demand, hub utilization rates, and charging temporal profiles, into decision-relevant economic metrics for operators, investors, and policymakers. The TCO formulation captures all major cost components over the system's lifetime, including upfront capital expenditures (CAPEX), operating and maintenance costs (OPEX), battery depreciation and replacement, electricity tariffs, and residual values, consistent with established e-mobility cost assessment practices[77,78].

Capital expenditures include vehicle acquisition, battery inventory (for swapping configurations), charging and swapping equipment, balance-of-system components, and installation costs. Operating expenditures include routine maintenance, battery servicing, electricity consumption under applicable tariff regimes, and station-level operational expenses. Battery depreciation is modeled explicitly using throughput-dependent degradation, reflecting the high-utilization duty cycles typical of boda-boda operations. Electricity costs are computed using both flat and time-of-use tariff structures, with charging energy allocated over time based on demand profiles derived from the travel-demand and charging-behavior models. All cost streams are evaluated using a discounted cash-flow approach over a five-year economic lifetime, applying a real discount rate of 10% to reflect prevailing local financing conditions and borrowing costs.

To account for uncertainty in key economic and operational parameters, the TCO analysis is complemented by a Monte Carlo simulation that varies electricity tariffs, battery lifetime, maintenance costs, and utilization rates within empirically plausible bounds[79]. This probabilistic approach estimates expected cost distributions and downside risk across infrastructure configurations, rather than relying solely on point estimates. The resulting metrics, including levelized cost per kilometer and net present cost, are used to compare swapping-only, charging-only, and hybrid configurations under consistent assumptions.

The techno-economic analysis is explicitly informed and validated by the financial randomized controlled trial (RCT), which serves as a behavioral-financial grounding layer rather than a direct input to

physical system models[80]. Empirical findings from the RCT constrain key financial assumptions, including repayment horizons, effective discount rates, and default risk, used in the Monte Carlo analysis[81]. Financing structures that show higher repayment stress or default probability in the RCT are treated as higher-risk scenarios when interpreting TCO outcomes, ensuring consistency between modeled financial feasibility and observed rider cash flow behavior. In this way, the RCT strengthens the credibility of the techno-economic results by anchoring them in real-world financing performance without overextending causal inference beyond the pilot scale.

By integrating engineering-derived energy demand, grid-aware operational constraints, and empirically informed financing assumptions, the techno-economic framework provides a coherent basis for comparing deployment pathways and identifying configurations that are not only cost-efficient in theory but also financially viable under the operating conditions of informal motorcycle taxi markets. The resulting TCO metrics feed directly into the grid-impact assessment and the subsequent macroeconomic analysis, ensuring continuity across modeling layers.

Mathematical formulation

1. Net present cost (NPC)

$$NPC = CAPEX + \sum_{t=1}^T \frac{OPEX_t + E_t + M_t + BD_t}{(1+r)^t}$$

Where:

- **CAPEX** = upfront cost of vehicle + battery + charging/swap equipment
- **OPEX_t** = annual operating expenses
- **E_t** = annual energy cost
- **M_t** = annual maintenance
- **BD_t** = annual battery depreciation
- **r** = 0.10 (discount rate)
- **T** = 5 years.

2. Battery depreciation

$$BD_t = C_{bat} \times \delta_t$$

With C_{bat} battery cost and δ_t annual degradation (10–12%).

3. Annual energy cost

$$E_t = (D[MULSGN]e) \times P_{ToU}$$

Where:

- **D** = annual distance (km),
- **e** = energy intensity (Wh/km),
- **P_{ToU}** = time-of-use tariff (off-peak or peak).

A comparative analysis was conducted to determine the cost per kilometer, payback period, and internal rate of return (IRR) for operators[82].

4. Levelized cost per kilometer (LCk)

$$LCk = \frac{NPC}{\sum_{t=1}^T \frac{D_t}{(1+r)^t}}$$

5. Payback period

$$PP = \min \left\{ t : \sum_{\tau=1}^t \frac{CF_{\tau}}{(1+r)^{\tau}} \geq CAPEX \right\}$$

6. Internal rate of return (IRR)

$$\sum_{t=1}^T \frac{CF_t}{(1+IRR)^t} - CAPEX = 0$$

7. Monte Carlo simulation

A 10,000-run Monte Carlo model perturbs

$$\{P_{ToU}, C_{bat}, \delta_t\}$$

using triangular distributions to estimate probability-based breakeven conditions for PAYGo and lease-to-own models[83].

This mathematical model supports a rigorous comparison of cost per kilometer, investment viability, and operator profitability across the three deployment strategies.

Distribution grid simulation

To evaluate the grid impacts of e-boda charging demand, distribution feeder simulations were performed using OpenDSS coupled with - QGIS-based spatial load allocation[84,85]. The modeled feeders represent high-density transit corridors, including Jinja Road, Bwaise-Kawempe, Masaka Road, Namasuba, Masanafu-Lugala, Nansana, and Entebbe Highway. Each feeder model includes transformer and line parameters sourced from ERA's 2024 grid inventory. Power-flow simulations were conducted under hourly charging profiles derived from the travel-demand model, testing for voltage deviation, thermal overload, and transformer utilization. Feeder reliability was assessed with a conservative threshold of <90% transformer utilization to ensure operational safety margins. Additional tests included smart charging and staggered swapping strategies, demonstrating that load management can reduce evening peak stress by up to 15%.

Financial pilot (randomized controlled trial)

A randomized controlled trial (RCT) was conducted as a pilot-scale financial feasibility and behavioral validation exercise to assess alternative financing structures for electric boda-boda adoption under real-world cash-flow constraints. Two financing schemes, lease-to-own (LTO) and pay-as-you-ride (PAYGo), were deployed among a sample of 120 riders in the Greater Kampala Metropolitan Area, with randomization stratified by Savings and Credit Cooperative (SACCO) membership to improve baseline comparability in access to credit and organizational support. The RCT was not designed to be population-representative but rather to empirically test the operational viability of financing mechanisms, repayment discipline, and short-term affordability dynamics relevant to informal transport operators.

Monthly repayment behavior, net cash-flow margins, downtime, and perceived affordability were tracked over 12 months using mobile payment logs and structured surveys. Treatment effects were estimated using an intent-to-treat difference-in-differences framework, enabling comparisons of financial performance across schemes while accounting for temporal shocks. The resulting evidence informs the plausibility of the financing assumptions used in the techno-economic analysis. It supports policy interpretation while explicitly acknowledging the limits of statistical generalization and inferences about city-wide adoption.

Macroeconomic assessment

The final analytical layer integrates micro-level engineering outcomes with macro-level economic structure[35]through an energy-oriented Input–Output (IO) model calibrated to Uganda's 42-sector supply–use tables. The model is hybridized with the KAMPALA-TIMES transportation module [4], enabling the translation of electrification scenarios into economy-wide effects. The KAMPALA-TIMES transportation module provides sector-specific changes in electricity demand, petroleum displacement, investment requirements, and vehicle-component supply, which become exogenous shocks to the IO system.

IO model structure

Let

- A = technical coefficient matrix (42×42),
- x = gross output vector,
- y = final demand vector,
- $L = (I - A)^{-1}$ = Leontief inverse.

Gross output is computed as:

$$x = L y$$

Policy shocks (e.g., renewable-powered swap hubs, tariff reform, import substitution) enter the model as perturbations to final demand:

$$\Delta x = L \Delta y$$

Energy and carbon intensities

Let

- e = energy-use coefficients (kWh or MJ per unit output),
- c = CO₂-emission coefficients (kg CO₂ per unit output).

Energy intensity of GDP:

$$EI = \frac{e'x}{GDP}$$

CO₂ intensity of GDP:

$$CI = \frac{c'x}{GDP}$$

Employment and welfare effects

Employment is evaluated using sectoral job coefficients w :

$$\Delta EMP = w' \Delta x$$

Household welfare is measured using Equivalent Variation (EV), where:

$$EV = \sum_h (U_h^{\text{shock}} - U_h^{\text{baseline}}) \cdot P_h$$

U_h = utility of household h ,

P_h = marginal utility of income.

TIMES-IO hybrid linkage

The integration of the KAMPALA-TIMES module with the energy-oriented Input–Output (IO) model is executed through a two-stage coupling mechanism that links micro-level engineering outcomes to macroeconomic system responses. In the first stage, engineering outputs from the KAMPALA-TIMES transportation module are translated into sector-specific economic shocks. Reductions in petrol consumption are mapped as contractions in the petroleum supply sector, while increased electricity demand stemming from charging and swapping operations is reflected as expanded output requirements for electricity generation, transmission, and distribution. Similarly, learning effects in local motorcycle and battery assembly are treated as positive shocks to the domestic manufacturing sectors, and the integration of PV–BESS systems is captured as increased demand for renewable-energy equipment and associated services.

In the second stage, these exogenous shocks propagate through the wider economy via the IO model's Leontief inverse. This propagation reflects the interdependence of 42 production sectors (Table 2), transmitting initial changes in energy demand and component supply to upstream intermediate suppliers, labor markets, and ultimately household incomes. Through this linkage, the hybrid TIMES-IO framework provides a coherent mechanism for assessing how localized technical interventions in the e-mobility system generate economy-wide impacts

Table 2

The 42 Production Sectors Used in the IO Model.

No.	Production Sector	No.	Production Sector
1	Agriculture (food crops)	22	Construction materials
2	Agriculture (cash crops)	23	Motor vehicle assembly
3	Forestry	24	Electric two-wheeler assembly
4	Fishing	25	Battery manufacturing & refurbishment
5	Mining & quarrying	26	Petroleum refining & fuels
6	Food processing	27	Electricity generation
7	Beverage & tobacco	28	Electricity transmission
8	Textile & apparel	29	Electricity distribution
9	Leather & footwear	30	Solar PV manufacturing
10	Wood products	31	BESS manufacturing
11	Paper & printing	32	Transport services (road – passenger)
12	Chemicals	33	Transport services (road – freight)
13	Plastics & rubber	34	Financial services
14	Cement & non-metallic minerals	35	Insurance services
15	Basic metals	36	Telecommunications
16	Fabricated metal products	37	Real estate
17	Machinery & equipment	38	Public administration
18	Electrical equipment	39	Education services
19	Electronics	40	Health services
20	Other manufacturing	41	Household services
21	Construction	42	Waste management & recycling

across GDP, employment, energy intensity, and carbon intensity, thereby ensuring a consistent evaluation of both engineering feasibility and national development implications.

The hybrid TIMES-IO approach enables analysis of how engineering changes, such as optimized hub siting, tariff reform, or battery-swapping penetration, spread through Uganda's production network, thereby impacting GDP, employment, energy intensity, and carbon intensity. This macroeconomic perspective ensures that electrification strategies are assessed not only for their technical feasibility but also for their contribution to national development and climate goals.

Model integration and validation

The analytical framework is implemented as a sequential, data-driven modeling pipeline in which outputs from each stage are systematically propagated as inputs to subsequent stages, ensuring internal consistency and transparency. Travel-demand modeling produces empirically derived daily distance distributions, temporal energy-demand profiles, dwell times, and spatial demand density maps, which directly inform the geospatial optimization of battery-swapping and charging hub locations and capacities under feeder-level constraints. The resulting hub configurations, together with hourly demand profiles and tariff schedules, are then passed to the techno-economic analysis to estimate total cost of ownership and operational cost metrics. These outputs are subsequently used to construct hourly load vectors for feeder-level distribution grid simulations, enabling assessment of transformer utilization, voltage profiles, and peak-load impacts under alternative deployment and demand-side management scenarios. Financial assumptions underlying the techno-economic scenarios are validated through a randomized controlled trial, which provides empirical evidence on repayment behavior and affordability without reparameterizing the physical system models. Finally, electricity demand changes, fuel displacement, and investment flows derived from the preceding stages are introduced into the energy-oriented input–output model to quantify macroeconomic effects.

Model validation occurs at multiple levels. Distribution-level load profiles are cross-checked against 2024 feeder statistics from the Electricity Regulatory Authority; cost parameters are verified against current OEM quotations and market data; and the input–output model is calibrated using the 2024 Uganda Bureau of Statistics social accounting matrix. Consistent variable definitions, explicit cross-referencing of

inputs and outputs, and harmonized notation across sections support traceability and reproducibility. Together, these integration and validation steps ensure that the results emerge from a coherent, end-to-end analytical pipeline rather than from loosely connected model components.

Sensitivity and robustness analysis

To evaluate the robustness of system-level conclusions under realistic operating uncertainty, a structured sensitivity analysis was conducted across key physical and operational parameters governing electric motorcycle energy consumption and infrastructure performance in the Greater Kampala Metropolitan Area (GKMA). Rather than relying on a single deterministic energy-intensity assumption, the analysis explores a plausible operating envelope of 35–65 Wh km⁻¹, reflecting observed variability in rider mass, payload, pavement condition, topography, and traffic dynamics in Kampala. Perturbations include ± 25 kg variations in combined rider–passenger mass, ± 30 kg payload changes associated with mixed passenger–cargo use, rolling resistance coefficients for good asphalt, degraded pavement, and mixed-surface operation, and road gradients up to $\pm 4\%$, consistent with localized elevation changes across major corridors.

For each perturbation set, impacts were propagated through the integrated modeling pipeline to quantify changes in daily energy demand per rider, total cost of ownership (TCO), leveled cost per kilometer (LCK), and feeder-level peak loading in grid simulations. Although individual vehicle energy demand varies meaningfully across the sensitivity envelope, aggregate system outcomes remain stable. In particular, hybrid swap–charge configurations consistently outperform charge-only systems in cost efficiency and grid impact across all tested scenarios. Feeder utilization remains within operational safety margins under coordinated demand-side management, and relative TCO rankings are preserved despite shifts in absolute cost levels. These results demonstrate that the study's core conclusions are not artifacts of narrow parameter choices but are robust to realistic uncertainty in operating conditions, reinforcing the credibility and transferability of the proposed grid-aware electrification pathways.

Results, analysis, and discussion

This section synthesizes results from the six-layer methodology: travel-demand analytics, geospatial siting, techno-economics, feeder simulations, the finance RCT, and the energy-oriented input–output model, followed by an integrated discussion within the context of current literature. Unless otherwise stated, confidence intervals (CIs) are calculated from 1000 bootstrap resamples; feeder safety margins aim for $\leq 90\%$ transformer utilization.

Travel-demand and energy profiles

Analysis of anonymized GPS trajectories collected from approximately 300–500 boda-boda motorcycles over a continuous 30-day period reveals a consistent and highly structured urban mobility pattern in the Greater Kampala Metropolitan Area (GKMA). Aggregated temporal profiles exhibit a pronounced trip-peaked structure, with dominant morning (06:30–09:00) and evening (16:30–20:00) peaks corresponding to commuter travel, and a secondary midday peak associated with parcel delivery, errands, and mixed passenger–cargo activity. This temporal structure is robust across weekdays and weekends and is consistent with observed informal transport dynamics in rapidly urbanizing African cities.

After quality filtering and map-matching, riders travel a mean of 92.4 km per day (95% CI: 86.1–98.8), completing an average of 26.7 trips per day (95% CI: 24.1–29.2). These estimates align with independent survey-based fleet statistics reported by local transport authorities and industry operators, providing external validation of the travel-

demand model. Spatial clustering analysis identifies persistent dwell and boarding hotspots around major taxi parks, trading centers, and *peri-urban* markets, including Bwaise, Nateete, Namasuba, Kabalagala, Najeera, Nansana, Masanafu–Lugala, Kajjansi, and Kireka, reflecting the strong coupling between informal mobility demand and commercial land-use patterns. Heatmaps show that approximately 80–82% of observed boarding and alighting events occur within a 1 km radius of ten dominant mobility hubs, underscoring the spatial concentration of demand along Kampala's primary corridors (Table 3).

Energy demand estimates are derived from the engineering energy model described in Section 3 rather than from a fixed-intensity assumption. Across the realistic operating envelope explored (35–65 Wh km⁻¹), daily traction energy requirements range from 3.2 to 6.0 kWh per rider, with a central tendency of 4.35 kWh day⁻¹ under mid-range operating conditions. Fig. 4 shows the aggregated hourly electricity demand curves, which show strong temporal alignment between mobility peaks and potential charging demand in the absence of demand-side management. These demand curves serve as direct input to feeder-level grid simulations and underscore the importance of coordinated charging and battery-swapping strategies.

Comparative analysis indicates that GKMA riders travel approximately 10–15% farther daily than reported for Kigali or Nairobi, consistent with findings from recent African e-mobility studies [33,86]. This spatial dominance and modal-substitution role of boda-bodas in Kampala align with Saddier's detailed analysis of the city's mobility system, which shows that motorcycle taxis function not merely as feeders but also as direct competitors to collective public transportation [34]. This higher duty, driven by combined passenger and courier functions, congestion-related detours, and dispersed *peri-urban* demand, results in greater cumulative energy throughput per vehicle. From a systems perspective, this strengthens the techno-economic case for battery-swapping configurations, which minimize downtime and preserve revenue-generating hours compared with plug-in charging under high utilization [87].

Geospatial siting and access – grid trade-offs

The multi-objective geospatial optimization model identifies 6 to 10 feasible battery-swapping and charging hubs across the Greater Kampala Metropolitan Area (GKMA), depending on infrastructure configuration and feeder constraints. The optimization explicitly allocates the hourly demand profiles derived in Fig. 4 to candidate hub locations by spatially distributing rider demand density and temporally assigning electricity demand to local feeders according to observed activity patterns. Under this integrated formulation, the hybrid configuration, consisting of six battery-swapping hubs and three slow-charging depots, achieves the lowest composite objective value by jointly minimizing rider access distance and feeder loading.

This configuration reduces the mean rider-to-hub distance to 0.84 km while maintaining maximum feeder utilization at 83% and staying within thermal and voltage stability margins across all peak periods

Table 3
Spatiotemporal Travel-Demand and Energy Profile Statistics.

Metric	Mean (95% CI)	Interpretation
Daily distance (km)	92.4 (86.1–98.8)	High-utilization, mixed passenger–cargo duty cycle
Trips per day	26.7 (24.1–29.2)	Short-haul dominance
Daily energy demand (kWh)	4.35 (3.2–6.0)	Sensitivity range: 35–65 Wh km ⁻¹
Peak-period share (%)	61.2	Morning and evening commuter peaks
Demand captured within 10 hubs (%)	81.7	Strong spatial concentration

Aggregated Hourly Electricity Demand Profile for Boda-Boda Operations (GKMA)

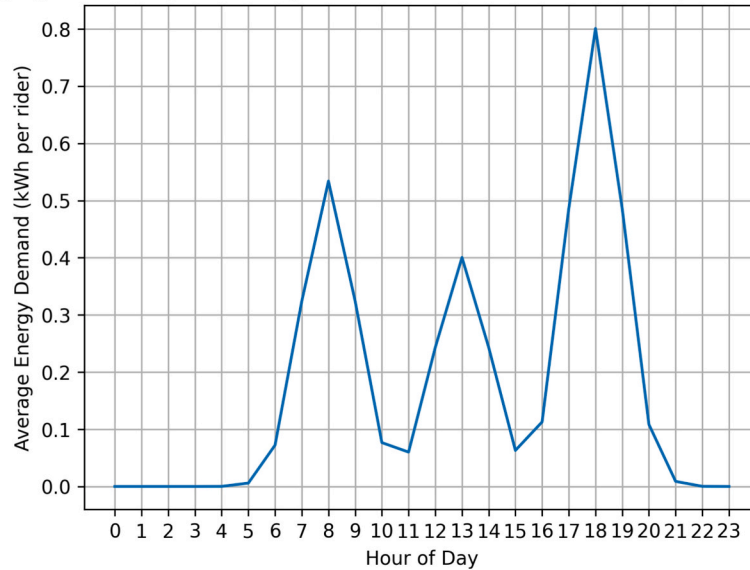


Fig. 4. Aggregated hourly electricity demand profile for boda-boda operations in GKMA. *Note:* The profile is derived from GPS-based travel-demand data and normalized to an average daily traction energy requirement of 4.35 kWh per rider. The curve shows pronounced morning (06:30–09:00) and evening (16:30–20:00) peaks, along with a secondary midday peak, reflecting commuter and mixed passenger–cargo activity patterns.

Table 4

Geospatial siting performance across infrastructure scenarios.

Scenario	Hubs (swap / charge)	Mean rider distance (km)	95th percentile distance (km)	Max feeder utilization (%)	Demand coverage within 1 km (%)
Charge-dominant	4 / 6	1.21	2.9	88	72
Swap-dominant	8 / 1	0.76	1.8	86	85
Hybrid (preferred)	6 / 3	0.84	2.0	83	82

(Table 4). Importantly, the highest hourly demand intervals identified in Fig. 4 are absorbed primarily by swap hubs along high-throughput corridors, thereby smoothing localized charging loads and limiting coincident demand on constrained feeders. In contrast, charge-dominant scenarios concentrate peak-period loads on fewer feeders, resulting in higher utilization and reduced spatial coverage.

Hybrid topologies, therefore, offer the most effective balance between accessibility and grid resilience. Swap-dominant systems minimize dwell time and preserve rider productivity, but they require higher capital investment. In contrast, charge-dominant layouts reduce upfront costs at the expense of longer idle times and greater curbside congestion. The hybrid solution mitigates these trade-offs by distributing temporal demand across a diversified infrastructure portfolio, maintaining grid performance while ensuring adequate spatial coverage. These findings align with Bhandari [88] and Alhazmi [23], who emphasize the role of modular swapping infrastructure in high-throughput urban corridors. By embedding feeder-capacity constraints and time-resolved demand allocation into the siting process, the analysis demonstrates the need for grid-aware planning to enable scalable, reliable electrification of informal motorcycle taxi systems.

Techno-economic performance (TCO)

The techno-economic assessment evaluates the levelized cost per kilometer (LCk) for charge-dominant, swap-dominant, and hybrid electrification configurations over a five-year planning horizon, discounted at a real rate of 10% to reflect prevailing capital-market conditions in Uganda. Cost inputs are based on contemporaneous quotations from local OEMs and battery-energy-storage suppliers. Electricity expenditures are calculated using the 2024 Electricity Regulatory Authority (ERA) time-of-use tariff schedule, with hourly energy

consumption derived directly from the demand profiles in Fig. 4. This integration ensures that cost outcomes reflect not only aggregate energy use but also the temporal alignment between charging behavior and tariff exposure.

Compared with a petrol baseline of USD 0.064–0.071 km^{−1} under 2024 pump prices, all electrification configurations deliver substantial cost reductions, with the hybrid configuration delivering the strongest performance. LCk values decline by 28–35%, driven by a balanced allocation of energy demand between fast battery swapping during peak operational hours and slower, tariff-optimized charging during off-peak periods. Swap-dominant systems achieve comparable cost reductions by minimizing rider downtime and preserving revenue hours, but incur higher battery leasing and station operating costs. In contrast, charge-dominant configurations exhibit higher LCk despite lower upfront infrastructure costs, primarily due to productivity losses from extended plug-in charging cycles and greater exposure to peak tariffs.

Sensitivity analysis shows that LCk increases by USD 0.003–0.005 km^{−1} under peak-heavy charging behavior, underscoring the importance of temporal load management. Conversely, off-peak charging and battery-buffered swapping reduce LCk by 6–9% through tariff arbitrage and lower effective grid-energy intensity (Table 5). These results

Table 5Total Cost of Ownership summary (USD km^{−1}).

Configuration	Energy & tariffs	Battery & lease	Station O&M	Vehicle O&M	LCk
Charge-dominant	0.017	0.012	0.005	0.013	0.047
Swap-dominant	0.015	0.016	0.007	0.011	0.049
Hybrid	0.016	0.013	0.006	0.011	0.046

demonstrate that tariff-responsive operational strategies, rather than static electricity pricing assumptions, are central to financial viability. The findings align with Prates et al. [89] on the role of shared infrastructure and battery leasing in LMIC contexts and slightly undercut LCK values reported in Nairobi pilots [90], reflecting Kampala's favorable off-peak tariffs and grid-aware hub siting.

Distribution grid impacts and smart-charging performance

Feeder-level distribution network simulations conducted in OpenDSS demonstrate that the cost-optimal hub configurations identified in Section 4.3, particularly the hybrid swap-plus-charge layout, are also technically feasible under realistic operating conditions. Hourly load vectors derived from the demand curves in Fig. 4 are spatially allocated to optimized hubs and mapped to their corresponding 11 kV and low-voltage feeders. When two coordinated operational strategies are jointly applied, namely (i) staggered battery-swapping intervals at 15-minute resolution and (ii) time-of-use (ToU), aligned charging schedules that shift depot charging to off-peak periods, transformer utilization on the most constrained feeders remains below 85%, consistent with ERA reliability and thermal loading guidelines.

Under baseline, uncoordinated charging scenarios, several feeders along high-demand corridors, including Jinja Road–CBD, Bwaise–Kawempe, Nansana, Masanafu, and Nateete, experience evening peak stresses due to the temporal coincidence of mobility demand and electricity consumption. Comparable feeder-level stress patterns have been reported in Nairobi under clustered electric-motorcycle charging, with Stratford et al. (2025) documenting localized transformer overloads from unmanaged e-boda charging, reinforcing the need for DSM-enabled rollout strategies in East African cities. The integration of demand-side management (DSM) measures reduces peak feeder demand by 14–17%, maintains minimum voltage above 0.96p.u., and prevents thermal overloads without requiring physical network reinforcement (Table 6). These improvements are most pronounced in the hybrid configuration, where battery swapping absorbs peak-hour mobility demand while bulk charging is deferred to off-peak windows.

The results indicate that Kampala's existing low- and medium-voltage distribution infrastructure can support large-scale e-boda electrification when grid-aware siting and temporal load management are implemented together. This finding aligns with DSM-based mitigation outcomes reported by Mohit & Pawar [91] and Roy and Ahmed [92], and it challenges prevailing assumptions that distribution network constraints are a binding barrier to electrification in African cities. Instead, the analysis shows that coordinated operational strategies, rather than immediate capital-intensive grid upgrades, offer a cost-effective pathway to maintaining system reliability while enabling rapid e-mobility deployment.

Finance pilot (RCT): affordability and repayment dynamics

A twelve-month randomized controlled trial (RCT) with 120 boda-boda riders evaluated the affordability and repayment performance of alternative financing mechanisms under real-world operating conditions. Participants were stratified by SACCO membership before randomization to ensure baseline comparability in income stability and access to informal credit. The trial compared lease-to-own (LTO) and pay-as-you-ride (PAYGo) schemes, focusing on cash-flow outcomes, repayment discipline, and operational continuity rather than

Table 7

Financial RCT outcomes (intent-to-treat analysis).

Outcome	Lease-to-own (mean)	PAYGo (mean)	Diff-in-Diff (β , p-value)
Monthly net cash margin (USD)	96.4	83.1	+11.7, p = 0.03
On-time repayment rate (%)	92.1	85.7	+5.8 pp, p = 0.04
Default probability (%)	6.5	10.7	−3.4 pp, p = 0.08
Idle time due to charging (hrs/week)	2.1	3.0	−0.8, p = 0.02
Willingness to recommend (0–10)	8.6	7.9	+0.7, p = 0.05

population-level adoption.

Results show that the LTO scheme consistently outperformed PAYGo across multiple financial and operational indicators (Table 7). Riders in the LTO arm achieved higher monthly net cash margins, demonstrated stronger on-time repayment behavior, and faced lower default risk. Qualitative follow-up surveys suggest that asset ownership strengthened rider commitment and enabled limited use of the motorcycle as collateral for minor repairs and short-term liquidity smoothing, thereby contributing to income stability. In contrast, PAYGo arrangements lowered entry barriers for riders with thin credit histories but were associated with narrower income buffers and moderately higher default probabilities.

Operational outcomes further reinforce these findings. LTO participants reported significantly lower idle time due to charging constraints, reflecting greater flexibility in charging and swapping behavior once ownership incentives were in place. These patterns align with broader regional evidence on asset-backed and hybrid fintech lending models in informal transport sectors[93]. Taken together, the RCT results support blended financing architectures that combine PAYGo accessibility with the repayment discipline and asset incentives of LTO schemes, particularly when anchored by SACCO-based guarantees to enable scalable and financially inclusive e-boda deployment.

Emissions, air quality, and MRV

By integrating grid-mix projections with empirically derived duty-cycle and charging profiles, the hybrid swap-charge scenario achieves an estimated 42% reduction in tailpipe CO₂ emissions and a 23–31% decline in localized PM_{2.5} concentrations at central loading bays and mobility hubs, relative to the internal combustion baseline. These reductions are evaluated over the use phase and across system operations and explicitly account for upstream electricity-generation emissions under Uganda's current grid mix. The results demonstrate that electric two-wheelers can deliver substantial climate and air-quality benefits in Sub-Saharan African cities, even where electricity supply remains partly thermal, provided that grid-aware siting, demand-side management, and tariff-responsive charging are implemented.

It is important to note that the emissions assessment focuses on operational and system-level impacts rather than the full cradle-to-grave life-cycle impacts of battery production, transport, and end-of-life management, which remain beyond the scope of this study. While comprehensive battery life-cycle assessment is critical for long-term sustainability evaluation, the present analysis is designed to inform near- to medium-term infrastructure planning, grid feasibility, and

Table 6

Feeder-level grid performance during peak periods.

Feeder	Baseline load (kVA)	Added EV load (kVA)	Utilization without DSM (%)	Utilization with DSM (%)	Minimum voltage (p.u., DSM)
Jinja Road–CBD	3,200	+410	92	84	0.96
Bwaise–Kawempe	2,450	+330	89	81	0.97
Entebbe Hwy–Kajjansi	2,900	+360	91	83	0.96

policy-aligned decarbonization pathways. Consistent with this framing, a GPS-enabled monitoring, reporting, and verification (MRV) architecture is developed to generate auditable CO₂ abatement datasets suitable for results-based finance and Article 6.2 mechanisms. Although PM_{2.5} reductions are modest relative to those in Asian megacities due to grid composition and dust resuspension [94,95], the associated public health co-benefits remain substantial and directly support Uganda's NDC objectives.

Macroeconomic feedbacks (TIMES–energy-oriented input-output model coupling)

Integrating the KAMPALA-TIMES energy systems module[3] with an energy-oriented input–output (E-IO) framework enables assessment of how grid-compatible e-boda electrification propagates through the broader economy via productivity, employment, and household income channels. Policy shocks representing concessional capital cost reductions, time-of-use (ToU) tariff alignment, and learning effects in local assembly are introduced consistently with techno-economic and grid outcomes from earlier sections. Crucially, financial parameters governing uptake and capital turnover, such as repayment horizons, default risk, and effective discounting, are constrained by empirical results from the randomized controlled trial (Section 4.5), ensuring that macroeconomic impacts reflect financing structures that are demonstrably viable under real cash-flow conditions.

Compared with the business-as-usual baseline, the model projects a 0.6% increase in GDP, driven by productivity gains from reduced downtime, lower operating costs, and reduced dependence on petroleum imports. Total employment expands by 1.8%, reflecting forward and backward linkages across motorcycle assembly, battery services, electrical work, fintech, and charging-infrastructure operations. Real household income for the lowest quintile rises by 1.2%, primarily due to lower transportation expenditures and increased participation in green service employment enabled by financially sustainable electrification pathways.

From an energy-system perspective, optimized hub siting and DSM limit system-wide electricity demand growth to 0.8% while reducing the fuel import bill by 2.4% [96]. These structural shifts yield a 1.9% reduction in the energy intensity of GDP and a 3.7% reduction in CO₂ intensity, confirming decoupling between economic output and emissions (Table 8). The results align with Uganda's Vision 2040 and NDC objectives and demonstrate that electrifying a high-utilization informal transport segment can generate economy-wide benefits disproportionate to its share of total energy demand. Consistent with E3 transition analyses[97,98], this study extends the literature by embedding two-wheeler electrification within a TIMES–E-IO hybrid framework, showing how engineering feasibility, tariff design, and empirically grounded financing jointly shape inclusive, low-carbon growth outcomes.

Table 8
Macroeconomic outcomes relative to baseline (2050).

Indicator	Value	Interpretation
GDP level change	+0.6%	Productivity gains and reduced fuel imports
Total employment	+1.8%	Job creation across assembly, services, and EVSE
Household real income (lowest quintile)	+1.2%	Transport cost savings and green job access
Fuel import bill	–2.4%	Petroleum displacement and forex savings
Electricity demand (system-wide)	+0.8%	Managed through DSM and ToU alignment
Energy intensity of GDP	–1.9%	Improved energy productivity
CO ₂ intensity of GDP	–3.7%	Structural decarbonization

The observed reductions in energy intensity (–1.9%) and CO₂ intensity (–3.7%) of GDP indicate emerging structural decoupling between economic growth and emissions, consistent with sustainable development pathways under the Paris Agreement. These gains demonstrate that electrifying even a single high-utilization segment, the two-wheeler fleet, advances Uganda's NDC objective of lowering emissions per unit of output while improving nationwide energy productivity. The decline in energy intensity supports SDG 7 on efficiency, whereas the reduction in CO₂ intensity directly reinforces SDG 13 on climate mitigation. Collectively, these results show that targeted urban transportation electrification can catalyze low-carbon, inclusive growth within constrained fiscal environments.

The integrated results demonstrate that Kampala's e-boda transition is driven primarily by strategic hub siting and demand-side management (DSM), rather than by increases in hardware capacity. System performance is governed by the spatial and temporal profiles of power withdrawal, extending Shibly et al. [96] by operationalizing a feeder-aware optimization framework suited to dense African CBDs. Economically, time efficiency and asset utilization outweigh nominal tariff levels; BESS-assisted swapping minimizes idle time and battery stress, reinforcing time as the principal determinant of last-mile electrification economics. Financial design also functions as a policy lever: randomized controlled trial evidence shows that SACCO-backed lease-to-own schemes broaden inclusion and sustain cash flow stability. When combined with optimized siting and DSM, electrification proceeds within feeder constraints, countering assumptions of LMIC grid fragility. Embedding MRV systems enables monetizable reductions in CO₂ and PM_{2.5}, positioning e-boda electrification as a cohesive engineering–finance–governance system for low-carbon urban mobility.

Conclusion

This study develops and empirically validates an integrated techno-economic, grid-aware framework for electrifying motorcycle taxi (boda-boda) fleets in the Greater Kampala Metropolitan Area (GKMA). By systematically linking GPS-based travel-demand characterization, geo-spatial infrastructure siting under feeder constraints, distribution-level power-flow simulation, total cost of ownership (TCO) analysis, pilot-scale financial experimentation, and an energy-oriented input–output macroeconomic model, the research establishes a coherent end-to-end analytical pipeline for evaluating informal two-wheeler electrification in constrained urban power systems.

Results show that a hybrid deployment strategy, combining battery swapping with tariff-optimized plug-in charging, achieves the most robust balance among cost efficiency, rider accessibility, and grid reliability. Across realistic operating and uncertainty ranges, the preferred configuration reduces levelized operating costs by 28–35%, keeps feeder utilization within regulatory thresholds through demand-side management, and supports high-utilization duty cycles without requiring near-term grid reinforcement. Evidence from the randomized controlled finance pilot indicates that SACCO-anchored lease-to-own schemes outperform pay-as-you-ride models in repayment discipline and cash-flow stability, underscoring the importance of financing design as a system enabler rather than an auxiliary consideration.

At the macroeconomic level, coupling the KAMPALA-TIMES model with an energy-oriented input–output framework shows that grid-compatible e-boda electrification yields positive economy-wide spillovers, including employment and household income gains, reduced fuel imports, and measurable declines in the energy and carbon intensities of GDP. Collectively, these findings indicate that electrification of high-utilization informal transport can be technically feasible, economically viable, and socially inclusive when pursued through coordinated infrastructure planning, tariff-responsive operations, and empirically grounded financial mechanisms.

Policy shocks arising from this study

1. Time-of-Use Tariff Reform: Introduce mandatory off-peak ToU tariffs for certified e-boda swap and charging hubs to incentivize load shifting and delay distribution network reinforcement.
2. Concessional Capital: Apply targeted CAPEX buy-downs or blended-finance instruments to battery-swapping infrastructure to accelerate scale-up while keeping feeder utilization below regulatory thresholds.
3. SACCO-Anchored Finance: Formalize SACCO-backed lease-to-own financing as a nationally endorsed mechanism for electrifying informal transport, leveraging demonstrated repayment discipline from the RCT.
4. Grid-Aware Siting Regulation: Mandate feeder-constrained geospatial siting assessments as a prerequisite for licensing high-throughput e-mobility infrastructure in urban areas.
5. MRV-Linked Results-Based Finance: Enable standardized MRV protocols for electric-motorcycle fleets to unlock results-based climate finance under Article 6.2 and in domestic carbon programs.

Future research frontiers and directions

Future research should build on this study by advancing the dynamic integration of urban mobility and electricity distribution systems, moving beyond static feeder assessments toward real-time, stochastic co-simulation of grid operations and mobility demand. Incorporating predictive load forecasting, probabilistic outage modeling, and high-resolution renewable intermittency into OpenDSS- or DigSILENT-based frameworks would enable more realistic evaluation of voltage stability, thermal limits, and demand-side management under high electric two-wheeler penetration. Parallel work is needed to establish robust battery circularity pathways through material flow analysis, second-life valuation, and end-of-life recycling models tailored to local regulatory and climatic conditions. In particular, extending the operational emissions and MRV framework presented in Section 4.6 to a full cradle-to-grave battery life-cycle assessment, including material extraction, manufacturing, second-life use, and end-of-life recycling, would provide a more comprehensive evaluation of the long-term environmental sustainability of battery-swapping systems. Socio-technical extensions should examine behavioral heterogeneity, equity dimensions, and cooperative-based financing mechanisms. At the same time, complete hybridization of energy-system optimization with recursive dynamic CGE models would allow endogenous feedbacks between income, technology choice, and industrial learning. Together, these research directions would strengthen the analytical foundations required for scalable, resilient, and climate-aligned e-mobility transitions in Sub-Saharan Africa.

Fig. 1 illustrates the integrated techno-economic-grid-policy triad underpinning the electrification of Kampala's boda-boda fleets through battery-swapping systems. At the core lies the electric boda-boda and battery-swap hub, symbolizing the operational interface between technology, users, and the power system. The left domain, Techno-Economic, captures the analytical foundation balancing charging versus swapping configurations, optimizing geospatial siting, and assessing total cost of ownership (TCO) under varying tariff and financing regimes. The right domain, Grid-Aware, emphasizes distribution-level constraints, incorporating feeder-capacity checks, power-flow simulations, and smart-siting algorithms to prevent localized overloading. The lower domain, Policy and Finance, bridges engineering insights with institutional levers, such as tariff reforms, concessional credit for SACCO-based operators, and carbon-accounting protocols for results-based financing. Their intersection represents a data-driven, investment-grade decision space that aligns technical feasibility with economic viability and policy coherence. Collectively, this triad ensures that e-boda electrification delivers triple dividends of reduced CO₂ emissions, enhanced rider livelihoods, and improved grid

resilience across the Greater Kampala Metropolitan Area (GKMA).

CRedit authorship contribution statement

Ismail Kimuli: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **John Baptist Kirabira:** Writing – review & editing, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Saadat L.K. Nakyejwe:** .

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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